

		Current Systems		Manual J			
	Sq Feet	Current Cooling – Tons	Current Heating	Cooling – BTU	Cooling – Tons	Cooling – Rounded Up	Probably do this ?
Bedroom	826	2.5		11,513	0.959	1	2
Downstairs	1736	3		45,113	3.759	4	5
Upstairs	2111	3.5		32,583	2.715	3	4



Project Summary

Entire House

GOODMAN MANUFACTURING, DAIKIN GROUP

Job: #3eP, Ahmad, I. 10727 Nw...
Date: Aug 15, 2024
By: Luis Rios

Project Information

For: Ahmad, Israr
10727 Nwport Bridge Lane Sugar Land, Tx 77498, Sugar Land, Tx 77498

Notes:

Design Information

Weather: Houston Hobby, TX, US

Winter Design Conditions

Outside db	37 °F
Inside db	70 °F
Design TD	33 °F

Ventilation Method MJ8

Heating Summary

Structure	44780 Btuh
Ducts (R-6.0)	19009 Btuh
Central vent (0 cfm)	0 Btuh

Humidification	0 Btuh
Piping	0 Btuh
Equipment load	63789 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

	Heating	Cooling
Area (ft²)	4673	4673
Volume (ft³)	45367	45367
Air changes/hour	0.28	0.15
Equiv. AVF (cfm)	212	113

Heating Equipment Summary

Make	n/a
Trade	n/a
Model	n/a
AHRI ref.	n/a

Efficiency	n/a
Heating input	
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	n/a

Summer Design Conditions

Outside db	94 °F
Inside db	75 °F
Design TD	19 °F
Daily range	L
Relative humidity	50 %
Moisture difference	50 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	54672 Btuh
Ducts (R-6.0)	26984 Btuh
Central vent (0 cfm)	0 Btuh

Blower 0 Btuh

Use manufacturer's data	n
Rate/swing multiplier	0.99
Equipment sensible load	81166 Btuh

Latent Cooling Equipment Load Sizing

Structure	8228 Btuh
Ducts	5861 Btuh
Central vent (0 cfm)	0 Btuh

Equipment latent load 14089 Btuh

Equipment Total Load (Sen+Lat)	95255 Btuh
Req. total capacity at 0.70 SHR	9.7 ton

Cooling Equipment Summary

Make	n/a
Trade	n/a
Cond	n/a
Coil	n/a
AHRI ref.	n/a

Efficiency	n/a
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0

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Page 1



Project Summary

First Floor

GOODMAN MANUFACTURING, DAIKIN GROUP

Job: #3eP, Ahmad, I. 10727 Nw...
Date: Aug 15, 2024
By: Luis Rios

Project Information

For: Ahmad, Israr
10727 Nwport Bridge Lane Sugar Land, Tx 77498, Sugar Land, Tx 77498

Notes:

Design Information

Weather: Houston Hobby, TX, US

Winter Design Conditions

Outside db	37 °F
Inside db	70 °F
Design TD	33 °F

Ventilation Method MJ8

Heating Summary

Structure	21375 Btuh
Ducts (R-6.0)	8529 Btuh
Central vent (0 cfm)	0 Btuh

Humidification	0 Btuh
Piping	0 Btuh
Equipment load	29905 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

	Heating	Cooling
Area (ft²)	1736	1736
Volume (ft³)	20337	20337
Air changes/hour	0.26	0.14
Equiv. AVF (cfm)	90	48

Heating Equipment Summary

Make
Trade
Model
AHRI ref

Efficiency	80 AFUE
Heating input	0 Btuh
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	1566 cfm
Air flow factor	0.052 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

Summer Design Conditions

Outside db	94 °F
Inside db	75 °F
Design TD	19 °F
Daily range	L
Relative humidity	50 %
Moisture difference	50 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	26403 Btuh
Ducts (R-6.0)	12298 Btuh
Central vent (0 cfm)	0 Btuh

Blower 0 Btuh

Use manufacturer's data	n
Rate/swing multiplier	0.99
Equipment sensible load	38469 Btuh

Latent Cooling Equipment Load Sizing

Structure	4019 Btuh
Ducts	2625 Btuh
Central vent (0 cfm)	0 Btuh

Equipment latent load 6644 Btuh

Equipment Total Load (Sen+Lat)	45113 Btuh
Req. total capacity at 0.70 SHR	4.6 ton

Cooling Equipment Summary

Make
Trade
Cond
Coil
AHRI ref

Efficiency	0 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	1566 cfm
Air flow factor	0.040 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.85

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Page 2



Project Summary Master GOODMAN MANUFACTURING, DAIKIN GROUP

Job: #3eP, Ahmad, I. 10727 Nw...
Date: Aug 15, 2024
By: Luis Rios

Project Information

For: Ahmad, Israr
10727 Nwport Bridge Lane Sugar Land, Tx 77498, Sugar Land, Tx 77498

Notes:

Design Information

Weather: Houston Hobby, TX, US

Winter Design Conditions

Outside db	37 °F
Inside db	70 °F
Design TD	33 °F

Ventilation Method MJ8

Heating Summary

Structure	8920 Btuh
Ducts (R-6.0)	2499 Btuh
Central vent (0 cfm)	0 Btuh

Humidification	0 Btuh
Piping	0 Btuh
Equipment load	11419 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

	Heating	Cooling
Area (ft²)	826	826
Volume (ft³)	7434	7434
Air changes/hour	0.27	0.14
Equiv. AVF (cfm)	33	18

Heating Equipment Summary

Make
Trade
Model
AHRI ref

Efficiency	80 AFUE
Heating input	0 Btuh
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	469 cfm
Air flow factor	0.041 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

Summer Design Conditions

Outside db	94 °F
Inside db	75 °F
Design TD	19 °F
Daily range	L
Relative humidity	50 %
Moisture difference	50 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	7952 Btuh
Ducts (R-6.0)	3631 Btuh
Central vent (0 cfm)	0 Btuh

Blower 0 Btuh

Use manufacturer's data	n
Rate/swing multiplier	0.99
Equipment sensible load	11513 Btuh

Latent Cooling Equipment Load Sizing

Structure	1005 Btuh
Ducts	765 Btuh
Central vent (0 cfm)	0 Btuh

Equipment latent load 1771 Btuh

Equipment Total Load (Sen+Lat)	13284 Btuh
Req. total capacity at 0.70 SHR	1.4 ton

Cooling Equipment Summary

Make
Trade
Cond
Coil
AHRI ref

Efficiency	0 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	469 cfm
Air flow factor	0.040 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.87

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Page 3



Project Summary

Second Floor

GOODMAN MANUFACTURING, DAIKIN GROUP

Job: #3eP, Ahmad, I. 10727 Nw...
Date: Aug 15, 2024
By: Luis Rios

Project Information

For: Ahmad, Israr
10727 Nwport Bridge Lane Sugar Land, Tx 77498, Sugar Land, Tx 77498

Notes:

Design Information

Weather: Houston Hobby, TX, US

Winter Design Conditions

Outside db	37 °F
Inside db	70 °F
Design TD	33 °F

Ventilation Method MJ8

Heating Summary

Structure	14485 Btuh
Ducts (R-6.0)	7981 Btuh
Central vent (0 cfm)	0 Btuh

Humidification	0 Btuh
Piping	0 Btuh
Equipment load	22465 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

	Heating	Cooling
Area (ft²)	2111	2111
Volume (ft³)	17596	17596
Air changes/hour	0.30	0.16
Equiv. AVF (cfm)	89	48

Heating Equipment Summary

Make
Trade
Model
AHRI ref

Efficiency	80 AFUE
Heating input	0 Btuh
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	1308 cfm
Air flow factor	0.058 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

Summer Design Conditions

Outside db	94 °F
Inside db	75 °F
Design TD	19 °F
Daily range	L
Relative humidity	50 %
Moisture difference	50 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	21259 Btuh
Ducts (R-6.0)	11520 Btuh
Central vent (0 cfm)	0 Btuh

Blower 0 Btuh

Use manufacturer's data	n
Rate/swing multiplier	0.99
Equipment sensible load	32583 Btuh

Latent Cooling Equipment Load Sizing

Structure	3205 Btuh
Ducts	2470 Btuh
Central vent (0 cfm)	0 Btuh

Equipment latent load 5675 Btuh

Equipment Total Load (Sen+Lat)	38257 Btuh
Req. total capacity at 0.70 SHR	3.9 ton

Cooling Equipment Summary

Make
Trade
Cond
Coil
AHRI ref

Efficiency	0 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	1308 cfm
Air flow factor	0.040 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.85

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Page 4